SAMANTHA (SAM) LUI

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EDUCATION

Bachelor of Computer Science (Third Year) | University of British Columbia *September 2024 – December 2027*
Advanced Diploma – Clinical Genetics (88%) | British Columbia Institute of Technology *August 2020 – October 2021*
Bachelor of Health Sciences (Dean's Honour Roll) | Simon Fraser University *September 2011 – December 2017*

PROFESSIONAL EXPERIENCE

UBC Faculty of Applied Sciences – Front-End Developer | Vancouver, BC *May 2026 – Present*

- Maintained and upgraded **9 Drupal 11** departmental websites using **DDEV** local, staging and deployment environments, delivering **25+** frontend enhancements (I.e., **Tailwind CSS**, module synchronization, SEO metadata optimization)
- Implemented a site-wide search feature with **Drupal Search API** and **Search 404**, mapping URL keywords against indexed content to redirect users to relevant search results
- Developed **3** custom **WordPress** websites for research groups and stakeholders, translating proof-of-concept designs into responsive, accessible web pages

CIHI – Applied Architecture Software Developer Intern | North York, ON *January 2026 – April 2026*

- Migrated 3 enterprise **Java** applications from Spring to **Spring Boot**, refactoring 20+ **Maven** dependencies and converting XML configurations to Java-based configs, improving cloud deployment stability on **Amazon EKS**
- Resolved containerization failures by replacing legacy **JSP** views with **Thymeleaf**, and automated **Docker** image builds with **Bamboo** CI/CD, reducing build and debugging time by **> 90%**
- Improved average test coverage across applications to **~88%**, using **Mockito** and **JUnit Jupiter** to improve regression confidence during Spring Boot migrations
- Contributed to an **AWS Bedrock proof-of-concept** to evaluate the usage of a multi-agent and human-in-the-loop validation workflow for automated logical data modelling

Shared Services Canada – I.T. Analyst Co-op Student | Vancouver, BC *August 2025 – December 2025*

- Developed an internal automation tool with **Python** and **Tkinter**, reducing ticket completion time by **75%** through Excel template auto-population and automated data entry
- Supported **Windows 11** endpoint deployment workflows and device management using **Microsoft Azure, MDM**, **MECM/SCCM**, and **PowerShell** for automated account provisioning
- Authored a **60+** page guide that was adopted as a core resource for student onboarding across regional support teams

TECHNICAL PROJECTS

PassPing | *React.js, Vite, TypeScript, Chrome APIs* | GitHub *March 2026 – July 2026*

- Developed a **Manifest V3** extension that delivers configurable U-Pass renewal reminders through persistent user preferences and background alarm scheduling, with automated builds and deployments executed through a **GitHub Actions CI/CD** pipeline

OverHacked! | *Phaser.js, Next.js, FastAPI, LangChain* | GitHub *January 2026*

- Built **FastAPI** backend endpoints, integrating OpenAI LLMs with **CrewAI** to dynamically generate programming quiz content for game interactions

JobQuest AI – AI-Powered Job App Assistant | *Next.js, AWS Bedrock, ElevenLabs* | GitHub *October 2025*

- Designed a gamified job search assistant with a **Next.js** video-game style UI, increasing engagement via interactive resume reviews and mock behavioural interviews with AI-powered features (**Bedrock**, **ElevenLabs**) for personalized insights

AniMori | *Electron.js, React, Vite, SQLite, Node.js, Jikan* | GitHub *August 2025 – December 2025*

- Developed a cross-platform **Electron.js** desktop anime-tracking app using **React** and **SQLite**, with automated metadata retrieval via the **Jikan API,** with **IPC** client-to-backend communication

TECHNICAL SKILLS

Languages | Python, Java, SQL, R, HTML5, CSS3, JavaScript
Frameworks/Libraries | React.js, Node.js/Express.js, Next.js, Electron.js, Spring Boot, JUnit, Mockito, FastAPI
Databases | PostgreSQL, MongoDB (NoSQL), SQLite, H2
Cloud & DevOps | GitHub Actions, Docker, JFrog Artifactory, Amazon Web Services (ECR, EKS)
Relevant Tools | GitLab, Bamboo, Jira, Postman, Vite, Jupyter, Selenium, Microsoft Azure (MDM, MECM/SCCM)